

Qualification Pack

Full Stack Development Associate

QP Code: NIE/SSC/Q0701

Version: 1.0

NSQF Level: 4

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NIE/SSC/Q0701: Full Stack Development Associate

Brief Job Description

The program enhances individuals' skills and promotes the responsible use of technology for broader socio-economic growth. After completing the course, learners are prepared to work as Full Stack Developers with a clear understanding of the entire website development process. They gain practical knowledge in client-side scripting, website maintenance, front-end and back-end development, and database integration. With additional experience, they may also handle server-side programming. A Web Developer combines coding, design, and functionality to build user-friendly websites, requiring both programming and basic graphic design skills.

Personal Attributes

The Full Stack Development Associate course encourages the development of key personal attributes essential for success in the IT industry. Learners are expected to demonstrate analytical thinking and problem-solving abilities to design effective and logical solutions. A strong attention to detail is crucial, particularly when writing and debugging code or designing user interfaces. Creativity plays a vital role in building visually appealing and user-friendly web applications. The course also nurtures adaptability and a continuous learning mindset, enabling learners to keep pace with evolving technologies. Effective communication, time management, and the ability to work collaboratively in teams are emphasized throughout the program. Additionally, learners are encouraged to uphold professional ethics, practice responsible use of technology, and exhibit self-motivation in completing tasks and contributing to real-world projects.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ITS/N0705: Design and Development of Front-End Application and managing and maintaining web source code--](#)
2. [NIE/ITS/N0702: Implementation of Backend Python programming \(functional and OOPs\)](#)
3. [NIE/ITS/N0703: Database connectivity with MySQL and Dynamic Web Application Development using Django and Front-end Technologies\(HTML/CSS/ JavaScripts/Bootstrap\)](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
5. [NIE/SSC/N0705: Implementation of Full Stack Development Associate](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	IT-ITeS
Sub-Sector	Software Development
Occupation	Web development, Web development
Country	India
NSQF Level	4
Credits	13
Aligned to NCO/ISCO/ISIC Code	4
Minimum Educational Qualification & Experience	12th grade Pass (12th or Equivalent) with NA of experience OR 10th grade pass (experience in IT or allied sector) with 3 Years of experience OR Not Applicable (Previous relevant Qualification of NSQF Level 3.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-04-IT-04172-2025-V2-NIELIT
NQR Version	2

Remarks:

NA

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NIE/ITS/N0705: Design and Development of Front-End Application and managing and maintaining web source code--

Description

This NOS focuses on developing skills to design and maintain front-end components of responsive websites. Learners will gain proficiency in using HTML5 for structuring content, CSS and Bootstrap for styling and layout, and JavaScript for interactivity and user input validation. It includes hands-on training with code editors and understanding key web concepts like URLs, IP addresses, and client-server architecture. The module also covers modern JavaScript (ES6+), DOM manipulation, and local storage. By the end, learners can build dynamic, user-friendly web interfaces following industry standards.

Scope

The scope covers the following :

- This NOS covers the skills required to design and develop static and responsive websites using HTML5, CSS, Bootstrap, and JavaScript. It includes understanding internet concepts, structuring web pages, applying styles and layouts, and adding interactivity through client-side scripting. The scope also includes using modern editors and ensuring basic form validations and responsiveness across devices.

Elements and Performance Criteria

Understand the Fundamentals of Internet and Web

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the concept of Internet, WWW, and their applications.
- PC2.** Identify types of website addresses, URLs, and IP addresses.
- PC3.** Differentiate between front-end and back-end.

Use Code Editors for Web Development

To be competent, the user/individual on the job must be able to:

- PC4.** Download and install free code editors (Notepad++, Sublime Text, VS Code).
- PC5.** Create, edit, and save HTML/CSS/JS files using editors.
- PC6.** Organize project folder structures.

Design Web Pages Using HTML

To be competent, the user/individual on the job must be able to:

- PC7.** Structure a basic HTML document with head and body sections.
- PC8.** Apply formatting tags (bold, italic, underline, etc.).
- PC9.** Insert images, paragraphs, comments, hyperlinks, and named anchors.

Style Web Pages Using CSS

To be competent, the user/individual on the job must be able to:

- PC10.** Identify and apply different types of CSS (inline, internal, external).
- PC11.** Use CSS selectors to target HTML elements.
- PC12.** Implement background, box, block, and text properties.

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Use Bootstrap Framework for Responsive Design

To be competent, the user/individual on the job must be able to:

PC13. Set up Bootstrap in a project.

PC14. Use Bootstrap grid system for responsive layouts.

PC15. Utilize built-in UI components (buttons, navbars, forms, etc.).

Apply Client-Side Interactivity Using JavaScript

To be competent, the user/individual on the job must be able to:

PC16. Declare variables and use data types in JavaScript.

PC17. Implement operators, control structures, and loops.

PC18. Create and call functions.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the basics of the Internet, websites, front-end/back-end structure, client-server scripting, and types of websites (static/dynamic).

KU2. Gain familiarity with popular code editors like Notepad, Sublime Text, and Visual Studio Code for creating and managing web files.

KU3. Learn to design structured web content using HTML and HTML5 elements, including forms, tables, multimedia tags, and semantic layout.

KU4. Apply styling and layout techniques using CSS and Bootstrap, including responsive design, UI components, and grid systems.

KU5. Develop interactive web pages using JavaScript by working with variables, functions, DOM manipulation, events, form validations, and ES6 features.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to clearly convey web concepts and use digital tools like browsers, editors, and online resources efficiently.

GS2. Skilled in HTML5, CSS, Bootstrap, and JavaScript for building responsive and interactive websites.

GS3. Demonstrates creativity in UI/UX design with attention to layout, accessibility, and user experience.

GS4. Capable of debugging code, handling errors, and applying logic to develop functional and well-structured websites.

GS5. Proficient with development editors follows coding standards, and applies modern web development practices.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand the Fundamentals of Internet and Web</i>	-	-	-	-
PC1. Explain the concept of Internet, WWW, and their applications.	-	-	-	-
PC2. Identify types of website addresses, URLs, and IP addresses.	-	-	-	-
PC3. Differentiate between front-end and back-end.	-	-	-	-
<i>Use Code Editors for Web Development</i>	-	-	-	-
PC4. Download and install free code editors (Notepad++, Sublime Text, VS Code).	-	-	-	-
PC5. Create, edit, and save HTML/CSS/JS files using editors.	-	-	-	-
PC6. Organize project folder structures.	-	-	-	-
<i>Design Web Pages Using HTML</i>	-	-	-	-
PC7. Structure a basic HTML document with head and body sections.	-	-	-	-
PC8. Apply formatting tags (bold, italic, underline, etc.).	-	-	-	-
PC9. Insert images, paragraphs, comments, hyperlinks, and named anchors.	-	-	-	-
<i>Style Web Pages Using CSS</i>	-	-	-	-
PC10. Identify and apply different types of CSS (inline, internal, external).	-	-	-	-
PC11. Use CSS selectors to target HTML elements.	-	-	-	-
PC12. Implement background, box, block, and text properties.	-	-	-	-
<i>Use Bootstrap Framework for Responsive Design</i>	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Set up Bootstrap in a project.	-	-	-	-
PC14. Use Bootstrap grid system for responsive layouts.	-	-	-	-
PC15. Utilize built-in UI components (buttons, navbars, forms, etc.).	-	-	-	-
<i>Apply Client-Side Interactivity Using JavaScript</i>	-	-	-	-
PC16. Declare variables and use data types in JavaScript.	-	-	-	-
PC17. Implement operators, control structures, and loops.	-	-	-	-
PC18. Create and call functions.	-	-	-	-
NOS Total	100	45	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0705
NOS Name	Design and Development of Front-End Application and managing and maintaining web source code--
Sector	Information Technology Sector
Sub-Sector	
Occupation	WEB DEVELOPMENT
NSQF Level	3
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

NIE/ITS/N0702: Implementation of Back End Web application using open source general-purpose scripting language and understand how its embedded into HTML

Description

Creating a National Occupational Standard (NOS) for the implementation of a back-end web application using an open-source general-purpose scripting language and understanding how it's embedded into HTML can help define the competencies required for this specific skill set. Below is a sample NOS description for this area:

Scope

The scope covers the following :

- Back End Development

Elements and Performance Criteria

Fundamental of Linux Operating System and Server-Side Development using PHP

To be competent, the user/individual on the job must be able to:

- PC1.** Able to create PHP scripts & code blocks
- PC2.** Use Expression, function & operator
- PC3.** PHP Syntax and embedding in HTML
- PC4.**
 - User Interface Design (UI) using HTML5
 - and PHP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Linux Operating System and its command
- KU2.** Programming in PHP

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental of Linux Operating System and Server-Side Development using PHP</i>	100	45	-	-
PC1. Able to create PHP scripts & code blocks	-	-	-	-
PC2. Use Expression, function & operator	-	-	-	-
PC3. PHP Syntax and embedding in HTML	-	-	-	-
PC4. • User Interface Design (UI) using HTML5 • and PHP	-	-	-	-
NOS Total	100	45	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0702
NOS Name	Implementation of Back End Web application using open source general-purpose scripting language and understand how its embedded into HTML
Sector	Information Technology Sector
Sub-Sector	Software Development
Occupation	Web development, Web development
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

NIE/ITS/N0702: Implementation of Backend Python programming (functional and OOPs)

Description

This course provides a comprehensive foundation in Python programming, beginning with its history, installation, and basic syntax. Learners are introduced to core programming concepts including data types, variables, operators, input/output operations, and control flow using conditional statements and loops. The course further explores Python's built-in functions, user-defined functions, and file handling techniques. Students gain hands-on experience with modular programming and the use of standard modules. A significant portion of the course is dedicated to Object-Oriented Programming (OOP), covering essential OOP principles like encapsulation, inheritance, polymorphism, abstraction, constructors, and exception handling. The syllabus is structured to build both logical thinking and practical coding skills for real-world applications.

Scope

The scope covers the following :

- This syllabus builds foundational and intermediate Python skills, including data types, operators, control structures, and functions. It covers file handling, modular programming, and Object-Oriented Programming concepts like inheritance and polymorphism. Learners also practice exception handling for writing robust code. The course prepares candidates for real-world programming and software development roles.

Elements and Performance Criteria

Set Up and Work with Python Environment

To be competent, the user/individual on the job must be able to:

- PC1.** Describe Python's background, features, and use cases.
- PC2.** Install Python on Windows/Linux and configure IDEs like Anaconda or VS Code.
- PC3.** Write and execute basic programs using proper syntax, variables, and indentation.

Apply Python Operators and Control Structures

To be competent, the user/individual on the job must be able to:

- PC4.** Use arithmetic, comparison, logical, assignment, and bitwise operators effectively.
- PC5.** Apply membership, identity, and ternary operators with correct precedence.
- PC6.** Implement conditional statements (if, if-else, elif) and loops (for, while, nested).

Work with Data Types and Collections

To be competent, the user/individual on the job must be able to:

- PC7.** Use built-in data types including numbers, strings, booleans, and None.
- PC8.** Create and manipulate lists, tuples, sets, dictionaries, and arrays.
- PC9.** Utilize relevant built-in methods for each data type.

Apply Control Structures

To be competent, the user/individual on the job must be able to:

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PC10. Implement decision-making structures such as if, else, elif, and nested conditions.

PC11. Use loops (for and while) to repeat tasks, including nested loops.

PC12. Control loop execution using break, continue, and pass statements.

Create and Use Functions

To be competent, the user/individual on the job must be able to:

PC13. Define and call user-defined functions with different types of parameters.

PC14. Utilize default and variable-length arguments.

PC15. Use Python's built-in functions and understand their utility.

Handle Files in Python

To be competent, the user/individual on the job must be able to:

PC16. Open files in various modes (read, write, append) and close them properly.

PC17. Read and write data to and from text files using relevant functions.

PC18. Use file-related methods such as open(), read(), readline(), and write().

Implement Object-Oriented Programming (OOP)

To be competent, the user/individual on the job must be able to:

PC19. Create and use classes and objects, defining attributes and methods.

PC20. Apply OOP principles like encapsulation, inheritance, polymorphism, and abstraction.

PC21. Use constructors, destructors, super(), and method overriding.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the fundamentals of Python, including its history, installation, syntax, and basic input/output operations.

KU2. Learn to implement control structures using conditional statements and loops, along with various function types and scopes.

KU3. Acquire skills in file handling, including reading, writing, and managing files using built-in functions

KU4. Understand modular programming concepts and how to use built-in and custom modules for better code organization.

KU5. Develop a strong foundation in Object-Oriented Programming, including classes, objects, inheritance, polymorphism, abstraction, and exception handling in Python.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to analyze problems logically and develop algorithmic solutions using conditional statements, loops, and functions in Python.

GS2. Skilled in writing clean, modular, and reusable code using core Python concepts, OOP principles, and built-in libraries.

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- GS3.** Proficient in using Python IDEs such as IDLE, Anaconda, and Visual Studio Code on both Windows and Linux platforms for development and debugging.
- GS4.** Capable of reading from, writing to, and processing data using Python file handling methods and working with built-in data structures like lists, sets, dictionaries, etc.
- GS5.** Applies best practices in modular programming, exception handling, and object-oriented design to build scalable, maintainable, and efficient applications.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Set Up and Work with Python Environment</i>	-	-	-	-
PC1. Describe Python's background, features, and use cases.	-	-	-	-
PC2. Install Python on Windows/Linux and configure IDEs like Anaconda or VS Code.	-	-	-	-
PC3. Write and execute basic programs using proper syntax, variables, and indentation.	-	-	-	-
<i>Apply Python Operators and Control Structures</i>	-	-	-	-
PC4. Use arithmetic, comparison, logical, assignment, and bitwise operators effectively.	-	-	-	-
PC5. Apply membership, identity, and ternary operators with correct precedence.	-	-	-	-
PC6. Implement conditional statements (if, if-else, elif) and loops (for, while, nested).	-	-	-	-
<i>Work with Data Types and Collections</i>	-	-	-	-
PC7. Use built-in data types including numbers, strings, booleans, and None.	-	-	-	-
PC8. Create and manipulate lists, tuples, sets, dictionaries, and arrays.	-	-	-	-
PC9. Utilize relevant built-in methods for each data type.	-	-	-	-
<i>Apply Control Structures</i>	-	-	-	-
PC10. Implement decision-making structures such as if, else, elif, and nested conditions.	-	-	-	-
PC11. Use loops (for and while) to repeat tasks, including nested loops.	-	-	-	-
PC12. Control loop execution using break, continue, and pass statements.	-	-	-	-
<i>Create and Use Functions</i>	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Define and call user-defined functions with different types of parameters.	-	-	-	-
PC14. Utilize default and variable-length arguments.	-	-	-	-
PC15. Use Python's built-in functions and understand their utility.	-	-	-	-
<i>Handle Files in Python</i>	-	-	-	-
PC16. Open files in various modes (read, write, append) and close them properly.	-	-	-	-
PC17. Read and write data to and from text files using relevant functions.	-	-	-	-
PC18. Use file-related methods such as open(), read(), readline(), and write().	-	-	-	-
<i>Implement Object-Oriented Programming (OOP)</i>	-	-	-	-
PC19. Create and use classes and objects, defining attributes and methods.	-	-	-	-
PC20. Apply OOP principles like encapsulation, inheritance, polymorphism, and abstraction.	-	-	-	-
PC21. Use constructors, destructors, super(), and method overriding.	-	-	-	-
NOS Total	50	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0702
NOS Name	Implementation of Backend Python programming (functional and OOPs)
Sector	Information Technology Sector
Sub-Sector	
Occupation	WEB DEVELOPMENT
NSQF Level	3
Credits	3
Version	1.0
Next Review Date	NA

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NIE/ITS/N0703: Database connectivity with PHP and MySQL

Description

Database connectivity with PHP and MySQL involves establishing a connection between PHP, a versatile server-side scripting language, and MySQL, a popular open-source relational database management system. This connection enables PHP scripts to interact with MySQL databases, allowing data retrieval, insertion, modification, and deletion, which is fundamental for dynamic web applications, content management systems, and e-commerce platforms. The PHP-MySQL combination provides the capability to create, query, and manage databases, allowing for the dynamic generation and manipulation of data-driven content on websites and web applications

Scope

The scope covers the following :

- Database connectivity

Elements and Performance Criteria

Learn the Configure PHP and MySQL

To be competent, the user/individual on the job must be able to:

- PC1.** Basic Programming using PHP
- PC2.** MYSQL to PHP Connectivity

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** PHP is a server-side scripting language primarily used for web development but also suitable for general-purpose programming. It is embedded within HTML and executed on the server, generating dynamic content that can be sent to the client's web browser. PHP is open-source, widely supported, and known for its ease of use and extensive community libraries and frameworks. It can interact with databases, handle form data, manage sessions, and perform a wide range of server-side tasks. PHP code is ex
- KU2.** MySQL is an open-source relational database management system (RDBMS) known for its speed, reliability, and scalability. It stores data in structured tables with relationships between them, making it an ideal choice for data-intensive applications. MySQL uses SQL (Structured Query Language) for querying and manipulating data. It supports advanced features such as transactions, indexing, and stored procedures. PHP can connect to MySQL databases using MySQLi (MySQL Improved) or PDO
- KU3.** Database connectivity with PHP and MySQL involves establishing a connection between a PHP script and a MySQL database server. This connection enables the PHP script to send SQL queries to the database, retrieve results, and perform various database operations. It is a fundamental aspect of web development, as it allows for the creation of dynamic web applications where data can be stored, retrieved, and manipulated in real-time. This connection is used for tasks such as user authentication, cont

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Learn the Configure PHP and MySQL</i>	30	15	-	-
PC1. Basic Programming using PHP	-	-	-	-
PC2. MYSQL to PHP Connectivity	-	-	-	-
NOS Total	30	15	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0703
NOS Name	Database connectivity with PHP and MySQL
Sector	Information Technology Sector
Sub-Sector	Software Development
Occupation	Web development, Web development
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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NIE/ITS/N0703: Database connectivity with MySQL and Dynamic Web Application Development using Django and Front-end Technologies(HTML/CSS/ JavaScripts/Bootstrap)

Description

This syllabus equips learners with the skills to develop, manage, and deploy secure, database-driven web applications using the Django framework and MySQL. It covers foundational knowledge of RDBMS, database connectivity, Django's MVT architecture, ORM, and user authentication. Learners gain hands-on experience in form handling, admin interface management, and integrating Bootstrap for responsive design. The scope includes project development, deployment practices, and implementing security features aligned with industry standards, preparing individuals for full-stack web development roles.

Scope

The scope covers the following :

- This syllabus equips learners with the skills to develop, manage, and deploy secure, database-driven web applications using the Django framework and MySQL. It covers foundational knowledge of RDBMS, database connectivity, Django's MVT architecture, ORM, and user authentication. Learners gain hands-on experience in form handling, admin interface management, and integrating Bootstrap for responsive design. The scope includes project development, deployment practices, and implementing security features aligned with industry standards, preparing individuals for full-stack web development roles.

Elements and Performance Criteria

Set Up Django Environment and Project Structure

To be competent, the user/individual on the job must be able to:

- PC1.** Install Django and create projects and apps using the command-line interface.
- PC2.** Understand the MVT architecture and role of manage.py, settings.py, and urls.py.
- PC3.** Configure URL routing and render views and templates correctly.

Design Front-End Using Templates and Static Files

To be competent, the user/individual on the job must be able to:

- PC4.** Use Django templates to display dynamic content and extend base layouts.
- PC5.** Include static assets (CSS, JS, images) using {% static %} and integrate Bootstrap.
- PC6.** Apply template tags like {% for %}, {% if %}, and pass data via context.

Handle Forms and User Input Securely

To be competent, the user/individual on the job must be able to:

- PC7.** Create and manage forms using HTML, Django Forms, and ModelForms.
- PC8.** Handle GET and POST requests, including CSRF token implementation.
- PC9.** Perform server-side validation and process user input safely

Build and Query Models Using Django ORM

To be competent, the user/individual on the job must be able to:

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PC10. Define models with fields and relationships; apply migrations to MySQL.

PC11. Use ORM queries like filter(), get(), exclude(), and order_by().

PC12. Add custom methods and use Meta options for model behavior.

Implement Views, Routing, and Navigation

To be competent, the user/individual on the job must be able to:

PC13. Develop function-based and class-based views (FBV, CBV).

PC14. Pass data from views to templates and capture URL parameters.

PC15. Use reverse() and redirect() for navigation control.

Manage Admin Interface and User Authentication

To be competent, the user/individual on the job must be able to:

PC16. Register and customize models in Django Admin with filters and search.

PC17. Create login, logout, and registration features using Django's auth system.

PC18. Apply decorators like login_required and manage roles and permissions

Secure and Deploy Django Applications

To be competent, the user/individual on the job must be able to:

PC19. Implement CSRF/XSS protection and SQL injection prevention via ORM.

PC20. Implement CSRF/XSS protection and SQL injection prevention via ORM.

PC21. Use Git for version control and understand deployment with WSGI/Gunicorn and LAMP stack.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the structure of Django projects, the MVT (Model-View-Template) architecture, project setup, URL routing, and view-template rendering for dynamic content generation.

KU2. Gain knowledge of creating and managing Django models, relationships, and queries using ORM, along with integrating MySQL databases and handling data migrations.

KU3. Learn how to create and validate forms, manage user input securely, and implement authentication, session handling, and access control using built-in Django tools.

KU4. Understand how to configure and connect Django with a MySQL database, apply migrations, and explore the basics of deploying Django apps using tools like WSGI/Gunicorn and Git in a production environment.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to build dynamic, database-driven web applications using Django, MVT architecture, and MySQL integration, including models, views, templates, and admin management.

GS2. Capable of designing scalable project structures, implementing secure form handling, managing authentication, and applying Django ORM for complex data queries and business logic.

GS3. Understanding of Git for source control, and familiarity with production deployment practices using WSGI/Gunicorn, LAMP stack, and environment configuration.

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GS4. Awareness and implementation of web security best practices including CSRF, XSS, SQL injection prevention, secure session handling, and user access control.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Set Up Django Environment and Project Structure</i>	-	-	-	-
PC1. Install Django and create projects and apps using the command-line interface.	-	-	-	-
PC2. Understand the MVT architecture and role of manage.py, settings.py, and urls.py.	-	-	-	-
PC3. Configure URL routing and render views and templates correctly.	-	-	-	-
<i>Design Front-End Using Templates and Static Files</i>	-	-	-	-
PC4. Use Django templates to display dynamic content and extend base layouts.	-	-	-	-
PC5. Include static assets (CSS, JS, images) using {% static %} and integrate Bootstrap.	-	-	-	-
PC6. Apply template tags like {% for %}, {% if %}, and pass data via context.	-	-	-	-
<i>Handle Forms and User Input Securely</i>	-	-	-	-
PC7. Create and manage forms using HTML, Django Forms, and ModelForms.	-	-	-	-
PC8. Handle GET and POST requests, including CSRF token implementation.	-	-	-	-
PC9. Perform server-side validation and process user input safely	-	-	-	-
<i>Build and Query Models Using Django ORM</i>	-	-	-	-
PC10. Define models with fields and relationships; apply migrations to MySQL.	-	-	-	-
PC11. Use ORM queries like filter(), get(), exclude(), and order_by().	-	-	-	-
PC12. Add custom methods and use Meta options for model behavior.	-	-	-	-
<i>Implement Views, Routing, and Navigation</i>	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Develop function-based and class-based views (FBV, CBV).	-	-	-	-
PC14. Pass data from views to templates and capture URL parameters.	-	-	-	-
PC15. Use reverse() and redirect() for navigation control.	-	-	-	-
<i>Manage Admin Interface and User Authentication</i>	-	-	-	-
PC16. Register and customize models in Django Admin with filters and search.	-	-	-	-
PC17. Create login, logout, and registration features using Django's auth system.	-	-	-	-
PC18. Apply decorators like login_required and manage roles and permissions	-	-	-	-
<i>Secure and Deploy Django Applications</i>	-	-	-	-
PC19. Implement CSRF/XSS protection and SQL injection prevention via ORM.	-	-	-	-
PC20. Implement CSRF/XSS protection and SQL injection prevention via ORM.	-	-	-	-
PC21. Use Git for version control and understand deployment with WSGI/Gunicorn and LAMP stack.	-	-	-	-
NOS Total	50	23	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0703
NOS Name	Database connectivity with MySQL and Dynamic Web Application Development using Django and Front-end Technologies(HTML/CSS/ JavaScripts/Bootstrap)
Sector	Information Technology Sector
Sub-Sector	
Occupation	WEB DEVELOPMENT
NSQF Level	3
Credits	3
Version	1.0
Next Review Date	NA

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

NIE/SSC/N0705: Implementation of Full Stack Development Associate

Description

This NOS enables learners to apply their full stack development skills through a hands-on project. They will build a complete web application using technologies like HTML, CSS, JavaScript, Bootstrap, Python, Django, and MySQL. The project includes stages such as planning, UI/UX design, coding, testing, and deployment. Learners will follow industry practices including version control with Git and secure coding standards. Through documentation and presentation, they demonstrate their ability to create dynamic and user-friendly applications aligned with real-world requirements.

Scope

The scope covers the following :

- The NOS scope for the Project module of the Full Stack Development Associate course involves the end-to-end development of a dynamic web application using both front-end and back-end technologies. Learners will apply their knowledge of HTML, CSS, JavaScript, Bootstrap, Python, Django, and MySQL to design and implement responsive and user-friendly web interfaces. The project includes form handling, database integration, authentication, and CRUD operations using Django ORM. Students will also gain exposure to version control with Git and basic deployment practices. The focus is on building real-time, data-driven applications that simulate industry requirements. This hands-on project prepares learners for real-world full stack development roles.

Elements and Performance Criteria

Project Planning and Requirement Analysis

To be competent, the user/individual on the job must be able to:

PC1. Identify a suitable real-world problem or business case for full-stack web development.

PC2. Define the functional and non-functional requirements of the project.

PC3. Create a detailed project plan including objectives, scope, timelines, technology stack, and resource needs.

Full Stack Web Application Development

To be competent, the user/individual on the job must be able to:

PC4. Design and implement the front-end using HTML, CSS, JavaScript, and Bootstrap to ensure responsive and interactive UI.

PC5. Develop the back-end functionality using Python and Django, incorporating CRUD operations, forms, and validation.

PC6. Integrate MySQL or SQLite databases using Django ORM and establish secure and scalable database interactions.

Testing, Deployment, and Documentation

To be competent, the user/individual on the job must be able to:

PC7. Perform unit testing, integration testing, and debugging to ensure application reliability and performance.

PC8. Prepare complete project documentation including system architecture, database schema, and version history.

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PC9. Deploy the web application on a local or cloud server and deliver a final presentation demonstrating project features and use cases.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the end-to-end process of full stack project development, including planning, designing, coding, testing, and deployment.
- KU2.** Gain knowledge of integrating front-end (HTML, CSS, JavaScript, Bootstrap) with back-end technologies like Python and Django.
- KU3.** Learn to design databases, perform CRUD operations using MySQL, and implement secure data handling using Django ORM.
- KU4.** Become familiar with development tools such as VS Code, Git, and deployment using servers like Gunicorn or NGINX.
- KU5.** Develop the ability to create responsive and user-friendly interfaces, manage user authentication, and prepare professional project documentation and presentations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to clearly convey ideas, collaborate with team members, and contribute actively to group projects and discussions.
- GS2.** Skilled in planning, organizing, and executing tasks efficiently within deadlines during project development.
- GS3.** Competent in front-end (HTML, CSS, JavaScript, Bootstrap) and back-end (Python, Django) development, along with database integration using MySQL/SQLite.
- GS4.** Capable of identifying issues, debugging code, and adapting to new tools and technologies in a fast-paced development environment.
- GS5.** Demonstrates use of version control tools like Git, applies web security best practices, and follows standard protocols for testing, documentation, and deployment.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Project Planning and Requirement Analysis</i>	-	-	-	-
PC1. Identify a suitable real-world problem or business case for full-stack web development.	-	-	-	-
PC2. Define the functional and non-functional requirements of the project.	-	-	-	-
PC3. Create a detailed project plan including objectives, scope, timelines, technology stack, and resource needs.	-	-	-	-
<i>Full Stack Web Application Development</i>	-	-	-	-
PC4. Design and implement the front-end using HTML, CSS, JavaScript, and Bootstrap to ensure responsive and interactive UI.	-	-	-	-
PC5. Develop the back-end functionality using Python and Django, incorporating CRUD operations, forms, and validation.	-	-	-	-
PC6. Integrate MySQL or SQLite databases using Django ORM and establish secure and scalable database interactions.	-	-	-	-
<i>Testing, Deployment, and Documentation</i>	-	-	-	-
PC7. Perform unit testing, integration testing, and debugging to ensure application reliability and performance.	-	-	-	-
PC8. Prepare complete project documentation including system architecture, database schema, and version history.	-	-	-	-
PC9. Deploy the web application on a local or cloud server and deliver a final presentation demonstrating project features and use cases.	-	-	-	-
NOS Total	-	-	30	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N0705
NOS Name	Implementation of Full Stack Development Associate
Sector	IT-ITeS
Sub-Sector	
Occupation	Web development
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of

Qualification Pack

NIELIT HQS.

2. The assessor assigns roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N0705.Design and Development of Front-End Application and managing and maintaining web source code--	100	45	-	-	145	39
NIE/ITS/N0702.Implementation of Backend Python programming (functional and OOPs)	50	22	-	-	72	19
NIE/ITS/N0703.Database connectivity with MySQL and Dynamic Web Application Development using Django and Front-end Technologies(HTML/CSS/JavaScripts/Bootstrap)	50	23	-	-	73	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	14
NIE/SSC/N0705.Implementation of Full Stack Development Associate	-	-	30	-	30	8
Total	220	120	30	-	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
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NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

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Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
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